



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics - Elementary Particles and

Astroparticles

Nicolò Favagrossa (68405A)

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UNIVERSITÀ
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DI MILANO



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Probing the EW Sector and BSM Physics with $W^\pm Zjj$

1 Introduction

- **Dataset:** 140 fb^{-1} of pp collisions at $\sqrt{s} = 13 \text{ TeV}$ (2015–2018).
- **Target:** Fully leptonic decays of $W^\pm Zjj$ production.
- **Measurements:** Integrated & differential fiducial cross-sections (σ_{EW} and σ_{strong}).
- **Goal:** Isolate Vector Boson Scattering (VBS).
- **New Physics:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).



The ATLAS Detector

1 Introduction

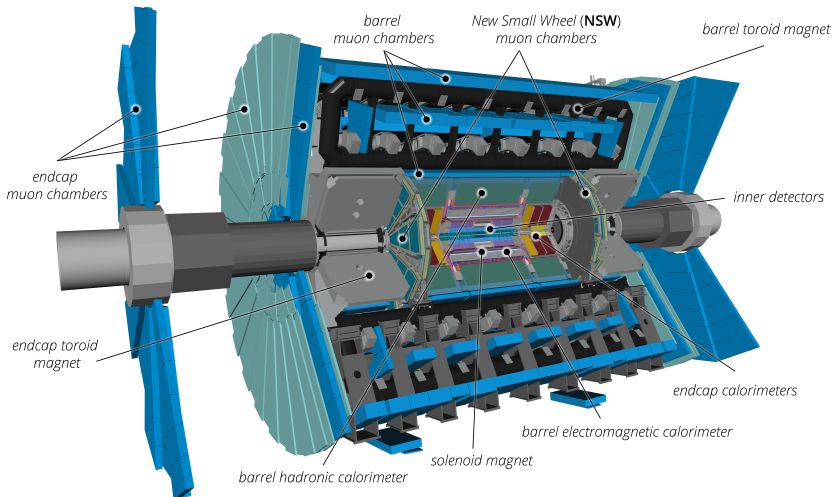




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Gauge Boson Self-Interactions

2 Theoretical Framework

- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry $SU(2)_L \times U(1)_Y$.
- Unlike QED, the generators of $SU(2)_L$ do not commute. As a consequence, the physical vector bosons ($W^\pm, Z, \gamma$) self-interact.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^\mu W_c^\nu (\partial_\mu W_{a\nu} - \partial_\nu W_{a\mu}) \quad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^\mu W_c^\nu W_{r\mu} W_{s\nu}$$

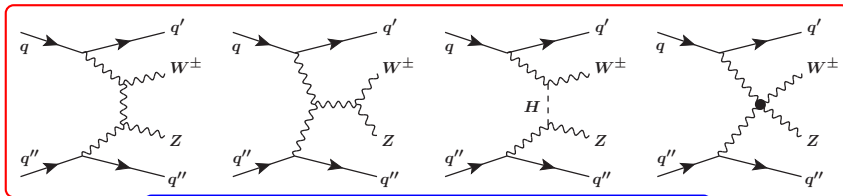
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from New Physics.



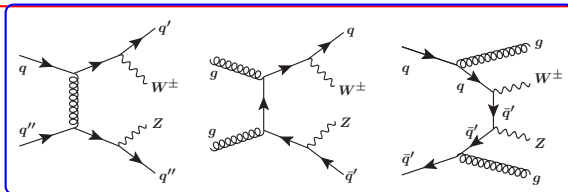
Experimental Signature: Vector Boson Scattering

2 Theoretical Framework

Vector Boson Scattering (VBS) $VV \rightarrow VV$ with $V = W/Z/\gamma$, is a key process with which to probe the $SU(2)_L \times U(1)_Y$ gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of $W^\pm Zjj$.



$W^\pm Zjj$ -EW



$W^\pm Zjj$ -QCD



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The $W^\pm Zjj$ -EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured $W^\pm Zjj$ events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an EFT framework. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective Wilson coefficients as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$: are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$ e $f_j^{(8)}$: dimensionless couplings
- Λ : is the energy scale of the new processes



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The squared scattering amplitude of the effective field theory prediction for $W^\pm Z jj$ production can be written as:

$$\left| A_{\text{SM}} + \sum_i c_i A_i \right|^2 = |A_{\text{SM}}|^2 + \sum_i c_i 2 \text{Re}(A_{\text{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)} / \Lambda^4$, A_{SM} is the SM scattering amplitude;
- $2 \text{Re}(A_{\text{SM}}^* A_i)$ is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i c_i^2 |A_i|^2$ is the pure dimension-8 contribution, and $\sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$ is the amplitude of interferences between two dimension-8 operators, called cross terms.



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework



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Fiducial Phase Space for Leptons

3 Fiducial Phase Space and MC Simulations

The $W^\pm Zjj$ cross-sections are measured in a fiducial phase space. Leptons produced in the decay of a hadron, a τ -lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

Category	Kinematic Variable	Requirement
Z leptons	Transverse momentum ($p_T^{\ell_Z}$)	$> 15 \text{ GeV}$
W lepton	Transverse momentum ($p_T^{\ell_W}$)	$> 20 \text{ GeV}$
All leptons	Pseudorapidity ($ \eta_\ell $)	< 2.5
Z boson	Invariant mass window ($ m_{\ell^+\ell^-} - m_Z^{\text{PDG}} $)	$< 10 \text{ GeV}$
W boson	Transverse mass (m_T^W)	$> 30 \text{ GeV}$
Angular isolation	Distance between Z leptons ($\Delta R(\ell_{Z1}, \ell_{Z2})$)	> 0.2
Angular isolation	Distance between W and Z leptons ($\Delta R(\ell_W, \ell_Z)$)	> 0.3



Fiducial Phase Space for Jets

3 Fiducial Phase Space and MC Simulations

In addition, at least two jets are required, reconstructed using the anti- k_t algorithm with radius parameter $R = 0.4$. Hard processes with a b -quark in the initial or final states, such as tZj production, are not considered as part of the $W^\pm Zjj$ -EW events.

Category	Kinematic Variable	Requirement
All jets	Transverse momentum (p_T^j)	$> 40 \text{ GeV}$
All jets	Pseudorapidity ($ \eta_j $)	< 4.5
Angular isolation	Distance between jets and leptons ($\Delta R(j, \ell)$)	> 0.3
Tagging jets	Opposite hemispheres ($\eta_{j1} \cdot \eta_{j2}$)	< 0
Tagging jets	Invariant mass (m_{jj})	$> 500 \text{ GeV}$
Jet Veto		
b -quark jets	Absence of b -jets in initial/final state	$N_{b\text{-jet}} = 0$



How to eliminate leptons ambiguity?

3 Fiducial Phase Space and MC Simulations

- Leptons and final-state neutrons that do not originate from hadron or τ -lepton decays, are matched to the $W^\pm$ and Z boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the W and Z resonances:

$$P = \left| \frac{1}{m_{(\ell^+, \ell^-)}^2 - (m_Z^{\text{PDG}})^2 + i \Gamma_Z^{\text{PDG}} m_Z^{\text{PDG}}} \right|^2 \times \left| \frac{1}{m_{(\ell', \nu_{\ell'})}^2 - (m_W^{\text{PDG}})^2 + i \Gamma_W^{\text{PDG}} m_W^{\text{PDG}}} \right|^2$$



Monte Carlo Simulation of the Signal

3 Fiducial Phase Space and MC Simulations

The simulation of $W^{\pm}Zjj$ events was primarily generated using the MadGraph5_aMC@NL02.6.5 MC event generator, interfaced with Pythia8.240. The simulation was divided into distinct components:

- **Electroweak production $W^{\pm}Zjj$ – EW:** The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order α_{EW}^6).
- **QCD production $W^{\pm}Zjj$ – QCD:** Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference $W^{\pm}Zjj$ – INT:** Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.



Background and Detector Simulation

3 Fiducial Phase Space and MC Simulations

In addition to the $W^{\pm}Zjj$ – QCD production, there are other processes that can mimic the signal in the detector.

- **Backgrounds with *prompt* (real) leptons:** Processes such as the production of ZZ pairs, WW , tribosons (VVV), or top quarks associated with vector bosons ($t\bar{t}V$, tZj). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Estimated directly using data-driven techniques.

All generated MC events were passed through the ATLAS detector simulation, based on Geant4, and processed using the same reconstruction software used for the data..



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Event Selection: Physics Objects Definition

4 Event selection & Background estimation

Data Quality & Trigger:

- Only data with stable beams and fully operational sub-detectors.
- Single-lepton triggers used (Combined efficiency $\approx 99\%$ for offline-selected $W^\pm Zjj$ events).

Physics Objects Reconstruction:

- **Prompt Leptons** e, μ, ν (E_T^{miss}): Reconstructed combining ID tracks with ECAL clusters or MS tracks. Strict isolation and impact parameter criteria are applied to reject non-prompt background.
- **Hadronic Jets**: Clustered using the *anti- k_t* algorithm ($R = 0.4$).
- **b-tagging**: Deep-learning algorithms identify displaced vertices from b-hadron decays (crucial for top-quark veto).



Physics Objects: Leptons

4 Event selection & Background estimation

Object	Selection Criteria
Muons	<i>Medium</i> Identification (Inner Detector + Muon Spectrometer) $p_T > 15 \text{ GeV}$, $ \eta < 2.5$ Impact parameters: $ d_0/\sigma_{d_0} < 3.0$, $ z_0 \sin \theta < 0.5 \text{ mm}$ Isolation efficiency: 90 – 99% (depending on p_T)
Electrons	<i>Medium Likelihood</i> Identification $p_T > 15 \text{ GeV}$, $ \eta < 1.37$ or $1.52 < \eta < 2.47$ (crack region veto) Impact parameters: $ d_0/\sigma_{d_0} < 5.0$, $ z_0 \sin \theta < 0.5 \text{ mm}$ Isolation efficiency: 89 – 99% (depending on p_T)

*Note: Impact parameter cuts are crucial to select **prompt** leptons and reject non-prompt background from heavy-flavor decays.*



Physics Objects: Jets & E_T^{miss}

4 Event selection & Background estimation

Object		Selection Criteria
Jets		Clustered via <i>anti-k_t</i> algorithm ($R = 0.4$) $p_T > 25 \text{ GeV}$, $ \eta < 4.5$ Pile-up suppression: <i>tight</i> ($p_T < 60 \text{ GeV}$, $ \eta < 2.4$) and <i>fJVT loose</i> ($p_T < 120 \text{ GeV}$, $2.5 < \eta < 4.5$)
Missing (E_T^{miss})	Energy	Proxy for the undetected neutrino. Computed as the negative vector sum of all calibrated <i>hard</i> objects (leptons, jets) plus a track-based <i>soft</i> term.



Event Selection: Boson Reconstruction

4 Event selection & Background estimation

Category	Selection Criteria
Baseline	Exactly 3 lepton candidates (μ or e) Veto on a 4th <i>loose</i> lepton ($p_T > 5$ GeV) Trigger matching: at least 1 lepton with $p_T > 25/27$ GeV
Z Boson Candidate	Opposite-Sign Same-Flavor (OSSF) lepton pair $ m_{\ell\ell} - m_Z^{\text{PDG}} < 10$ GeV (If ambiguous, the pair closest to m_Z is chosen)
W Boson Candidate	The remaining 3rd lepton + E_T^{miss} Stricter requirements: <i>Tight</i> ID & Isolation, $p_T > 20$ GeV Transverse mass: $m_T^W > 30$ GeV



Signal Region (SR): VBS Topology

4 Event selection & Background estimation

Category	Selection Criteria
Tagging Jets	The two highest- p_T jets in the event ($p_T > 40$ GeV) Must reside in opposite hemispheres ($\eta_{j1} \cdot \eta_{j2} < 0$)
Kinematic Cuts	$m_{jj} > 150$ GeV (rejects triboson background) $m_{jj} > 500$ GeV (isolates the EWK VBS process from QCD)
Background Veto	Exact b-veto ($N_{b\text{-jet}} = 0$) (strongly suppresses top-quark background)



Background Estimation Strategy

4 Event selection & Background estimation

Two Main Categories:

- **Irreducible:** True prompt leptons mimicking the signal topology.
- **Reducible (Fakes):** Non-prompt leptons or hadronic jets misidentified as prompt leptons.

Estimation Techniques in the Simultaneous Fit:

- **Data-driven (Control Regions):** Used for dominant backgrounds. Extract normalization factors (μ_{bkg}) directly from data.
- **Pure Monte Carlo:** Used for rare processes. Constrained by theoretical uncertainties.
- **Matrix Method:** Fully data-driven approach for reducible fakes, bypassing MC fragmentation modelling.



Irreducible Backgrounds

4 Event selection & Background estimation

Constrained via Control Regions (ZZ-CR & b-CR):

- $ZZ \rightarrow 4\ell$ (ZZjj-QCD): Survives via lepton acceptance loss. Constrained in the orthogonal **ZZ-CR** (requiring a 4th *loose* lepton).
- **Top-quark** ($t\bar{t} + V, tZj$): Survives via b-tagging inefficiency. Constrained in the **b-CR** (inverting the b-veto, ≥ 1 b-jet).

Estimated via Pure Monte Carlo:

- **Tribosons** (VVV) & ZZjj-EW: Yields too low to define a dedicated CR.



Reducible Backgrounds (Fake Leptons)

4 Event selection & Background estimation

Sources:

- $Z + \text{jets}, Z\gamma, t\bar{t}$ without a true 3rd prompt lepton.
- Light/heavy-flavor jets decaying semi-leptonically or photon conversions.

The Matrix Method:

- Relies on two lepton identification criteria: *Loose* (baseline) and *Tight* (signal).
- Uses real lepton efficiency (ϵ) and fake rate (f) measured in independent data samples.
- Solves a linear system to estimate the fake contribution directly in the Signal Region.



Expected and Observed Numbers of Events Pre-Fit

4 Event selection & Background estimation

	SR ($N_{\text{jet}} = 2$)	SR ($N_{\text{jet}} \geq 3$)	b -CR	ZZ-CR
Data	169	477	666	210
Total pred.	231 ± 12	550 ± 50	660 ± 40	205 ± 11
$W^{\pm}Zjj$ — EW	65.0 ± 3.5	60 ± 6	4.82 ± 0.28	0.725 ± 0.014
$W^{\pm}Zjj$ — QCD	125 ± 9	380 ± 50	77 ± 18	6.2 ± 0.7
$W^{\pm}Zjj$ -INT	1.3 ± 0.6	5.3 ± 2.6	0.58 ± 0.29	0.22 ± 0.11
$t\bar{t} + V$	0.66 ± 0.04	20.2 ± 0.7	289 ± 10	9.89 ± 0.28
tZj	8.78 ± 0.34	19.7 ± 1.2	134 ± 4	0.432 ± 0.005
ZZ-QCD	9.6 ± 0.4	32.0 ± 2.5	10.1 ± 0.6	159 ± 9
ZZ-EW	2.2 ± 0.6	4.4 ± 1.1	0.25 ± 0.06	23 ± 6
VVV	0.41 ± 0.10	2.0 ± 0.5	0.39 ± 0.10	4.1 ± 1.1
Misid. leptons	18 ± 4	28 ± 7	150 ± 40	1.7 ± 0.5



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Signal Extraction: Multivariate Analysis (BDT)

5 Measurement methodology and results

The Challenge:

- $WZjj$ -EW (signal, $\mathcal{O}(\alpha_{EW}^6)$) and $WZjj$ -QCD (background, $\mathcal{O}(\alpha_s^2 \alpha_{EW}^4)$) share the exact same final state.
- Cut-based analysis is sub-optimal $\Rightarrow$ Boosted Decision Tree (TMVA).

BDT Input Variables (15 observables):

- *Jet Kinematics*: m_{jj} , $\Delta\eta_{jj}$, $\Delta\phi_{jj}$, jet multiplicity.
- *Boson Kinematics*: p_T^W , p_T^Z , m_T^{WZ} .
- *Mixed Correlations*: ℓ -centrality (ζ_{lep}), event balance.

BDT Output:

- **The Score**: A single output that allows you to differentiate the signal.

Note: A different BDT is also used to differential the background in the b -CR between $t\bar{t}V$ tZj .



The Global Profile Likelihood Fit

5 Measurement methodology and results

Simultaneous Extended Binned Negative Log-Likelihood:

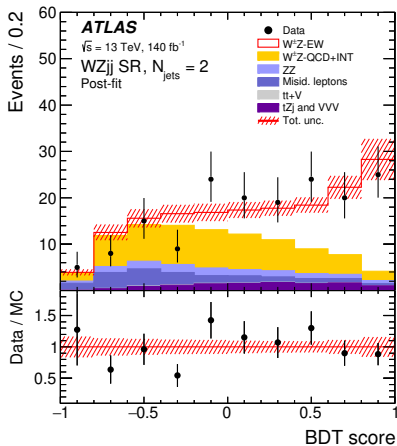
$$-\ln \mathcal{L}(\mu_{sig}, \mathbf{K}, \boldsymbol{\theta}) = \sum_r \sum_i \left(\mu^{[r,i]} - N^{[r,i]} \ln \mu^{[r,i]} \right) + \sum_j \frac{\theta_j^2}{2}$$

- **Regions (r):** SR ($N_{jets} = 2$), SR ($N_{jets} \geq 3$), b-CR, ZZ-CR.
- **Observables ($N^{[r,i]}$):** BDT score in SRs, m_{jj} in ZZ-CR, b-CR BDT in b-CR.
- **Parameters of Interest (POI):** Signal strengths μ_{sig} for EWK and QCD.
- **Unconstrained Parameters ($\mathbf{K}$):** Background normalizations ($K_{t\bar{t}V}$, K_{ZZ}) extracted dynamically from CRs.
- **Nuisance Parameters ($\boldsymbol{\theta}$):** Theoretical and experimental systematics, constrained by the Gaussian penalty term $\theta_j^2/2$.

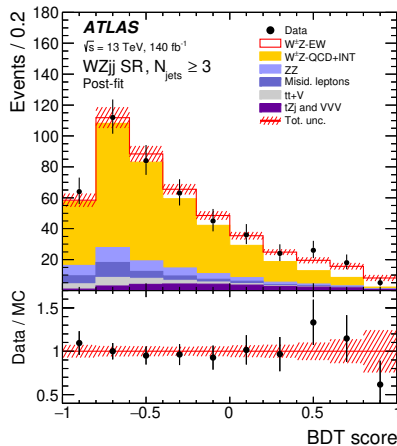


Post-fit distributions SR ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

5 Measurement methodology and results



(a)

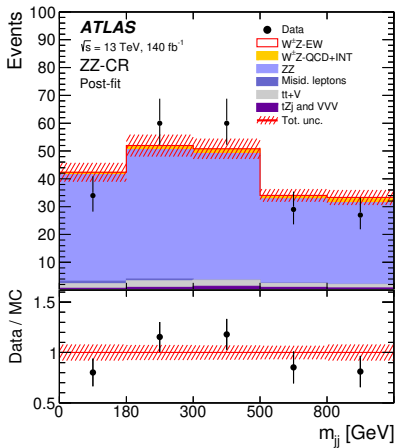


(b)

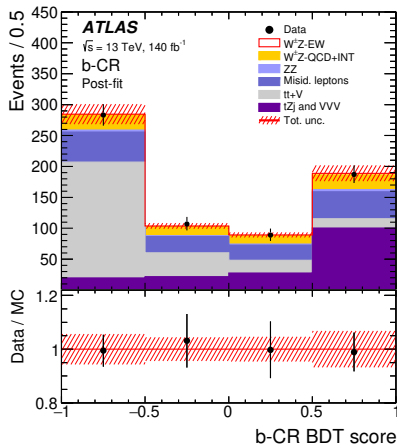


Post-fit distributions ZZ-CR, b-CR

5 Measurement methodology and results



(c)



(d)



Fiducial Cross-Section Extraction

5 Measurement methodology and results

The fiducial cross-sections are extracted directly from the profile-likelihood fit. The normalisation factors (μ_{EW} , μ_{QCD}) scale the theoretical MC predictions within the fiducial phase space:

$$\sigma_{WZjj-EW} = \sum_i \mu_i^{WZjj-EW} \cdot \sigma_{i,th.MC}^{WZjj-EW}$$
$$\sigma_{WZjj-strong} = \sum_i \left(\mu_i^{WZjj-QCD} \cdot \sigma_{i,th.MC}^{WZjj-QCD} + \mu_i^{WZjj-INT} \cdot \sigma_{i,th.MC}^{WZjj-INT} \right)$$

**Index i runs over the two SR categories: $N_{jets} = 2$ and $N_{jets} \geq 3$.*

- **Interference term:** μ_{INT} is constrained as $\sqrt{\mu_{EW}} \cdot \sqrt{\mu_{QCD}}$.



Results and Comparison with SM Predictions

5 Measurement methodology and results

VBS (EWK Production)

$$\sigma_{\text{obs}}^{\text{EWK}} = 0.368 \pm 0.037_{\text{stat}} \pm 0.059_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{EWK}} = 0.370 \pm 0.030 \text{ fb}$$

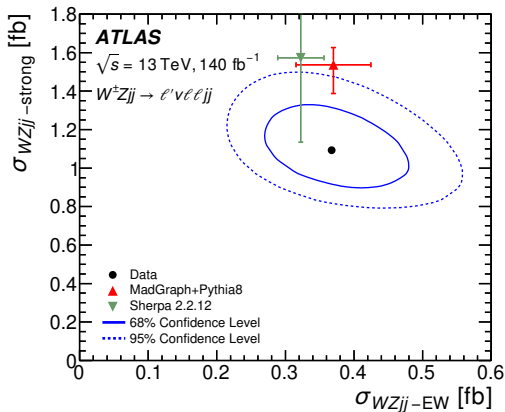
⇒ Excellent SM agreement

Strong + Int.

$$\sigma_{\text{obs}}^{\text{strong}} = 1.093 \pm 0.066_{\text{stat}} \pm 0.131_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{strong}} = 1.537 \pm 0.150 \text{ fb}$$

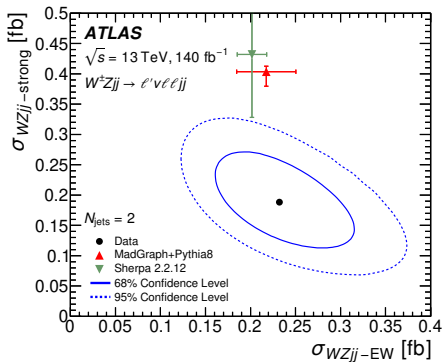
⇒ Deficit factor ~ 0.7 (1.8σ tension)



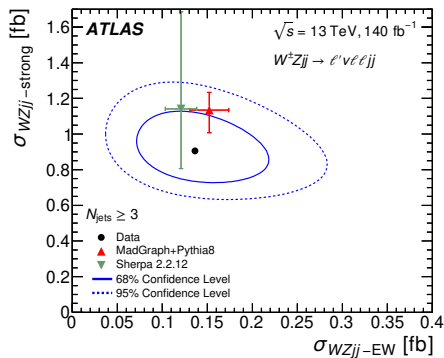


$W^\pm Zjj$ – EW and $W^\pm Zjj$ -strong ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

5 Measurement methodology and results



(a)

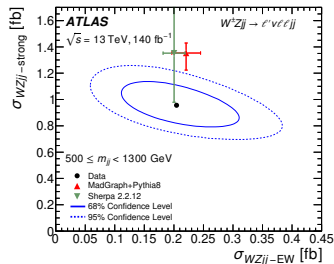


(b)

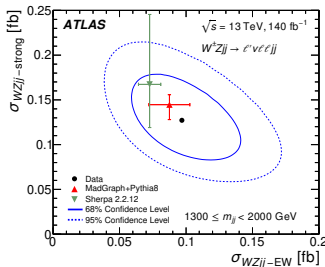


$W^{\pm}Zjj$ – EW and $W^{\pm}Zjj$ -strong in different m_{jj} bins

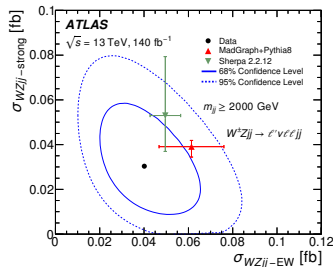
5 Measurement methodology and results



(a)



(b)



(c)



$W^{\pm}Zjj$ differential measurements

5 Measurement methodology and results

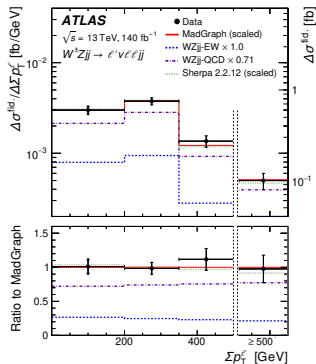
Differential measurements of $W^{\pm}Zjj$ production cross-section are performed as a function of three variables sensitive to anomalies in the quartic gauge coupling in $W^{\pm}Zjj$ events:

- the scalar sum of the trasverse momenta of the three charged leptons associated with W and Z bosons $\sum p_{\text{T}}^{\ell}$;
- the difference in azimuthal angle $\Delta\phi(W, Z)$ between the W and Z bosons' directions;
- the trasverse mass of the $W^{\pm}Z$ system m_{T}^{WZ} .

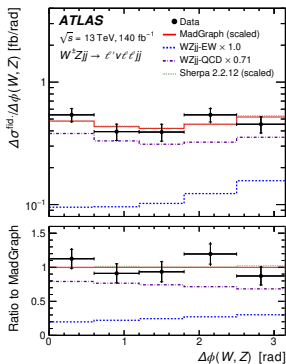


$W^\pm Z jj$ differential measurements

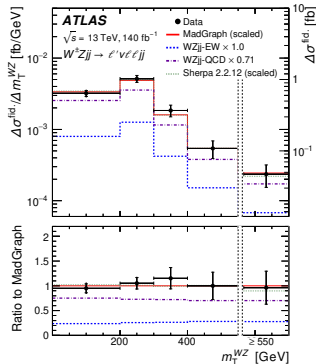
5 Measurement methodology and results



(a)



(b)



(c)



Search for aQGC: Methodology

Anomalous Quartic Gauge Couplings (aQGC) modeled via EFT dimension-8 operators drastically enhance the cross-section at high energy scales ($\hat{s}$).

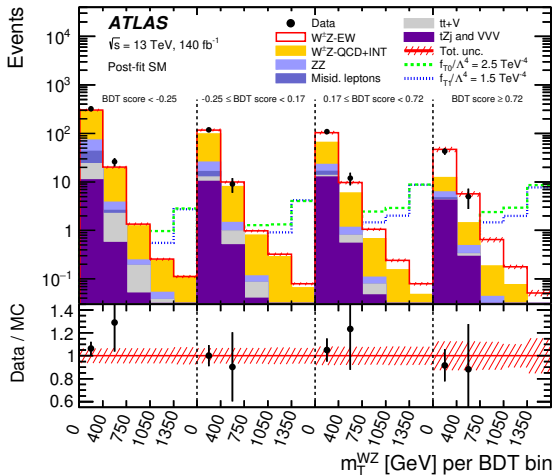
The 2D Fit Strategy

- **Observables:** The search exploits a 2D distribution combining the BDT score (to separate EWK from QCD) and the transverse mass m_T^{WZ} (sensitive to $\hat{s}$).
- **Binning:** 4 Bins in BDT score $\times$ 5 bins in $m_T^{WZ} \rightarrow$ Unrolled into a 1D histogram of 20 statistically independent bins.
- **Test Statistic:** Profiled Likelihood Ratio $q(\vec{c})$ to extract 95% Confidence Level (CL) limits on the Wilson coefficients $f_j^{(8)} / \Lambda^4$.



2D BDT vs m_T^{WZ} Distribution for EFT Search

5 Measurement methodology and results





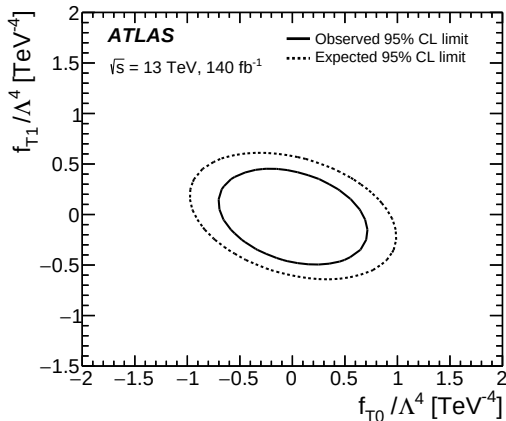
EFT Limits (Without Unitarization)

5 Measurement methodology and results

1D Limits (95% CL)

Coupling	Exp. [TeV^{-4}]	Obs. [TeV^{-4}]
f_{T0}/Λ^4	$[-0.80, 0.80]$	$[-0.57, 0.56]$
f_{T1}/Λ^4	$[-0.52, 0.49]$	$[-0.39, 0.35]$
f_{T2}/Λ^4	$[-1.6, 1.4]$	$[-1.2, 1.0]$
f_{M0}/Λ^4	$[-8.3, 8.3]$	$[-5.8, 5.6]$
f_{M1}/Λ^4	$[-12.3, 12.2]$	$[-8.6, 8.5]$
f_{M7}/Λ^4	$[-16.2, 16.2]$	$[-11.3, 11.3]$
f_{S02}/Λ^4	$[-14.2, 14.2]$	$[-10.4, 10.4]$
f_{S1}/Λ^4	$[-42, 41]$	$[-30, 30]$

2D Contour (f_{T0} vs f_{T1})





Tree-Level Unitarity Violation and Clipping

5 Measurement methodology and results

The Unitarity Bound

The EFT is not a complete model. Non-zero dimension-8 operators violate tree-level unitarity at high energies.

The "Clipping" Technique

- EFT contributions are removed above a cut-off scale E_c ($m_{WZ} > E_c$).
- Replaced by SM predictions to restore unitarity.

Unitarized Limits (at crossing bound):

- $f_{T0}/\Lambda^4 \in [-1.5, 1.6] \text{ TeV}^{-4}$
- $f_{T1}/\Lambda^4 \in [-0.7, 0.6] \text{ TeV}^{-4}$

Evolution of 95% CL limits vs cut-off scale $\Rightarrow$

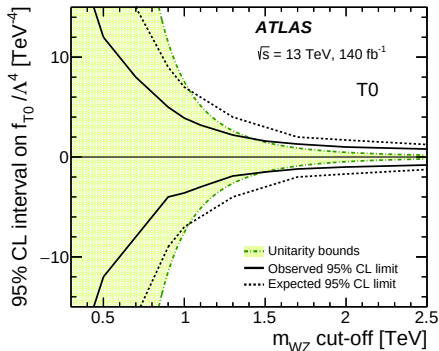




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Conclusions: VBS as a Precision Probe for the EWK Sector

6 Conclusion

From Discovery to Precision Measurement (Run 2, 140 fb^{-1})

- **Integrated cross-section:** $\sigma_{WZjj\text{-EW}} = 0.368 \pm 0.037 \text{ (stat)} \pm 0.059 \text{ (syst) fb.}$
- **Differential measurements** (via Bayesian Unfolding) on aQGC-sensitive variables: $m_{jj}, \Delta\phi, m_T^{WZ}, \sum p_T^\ell.$

Standard Model Verification and Modelling Challenges

- **EWK Production:** Excellent Data/Theory agreement in the Signal Region.
- **QCD Background:** Tension of 1.8σ (Data $\sim 30\%$ lower than nominal MadGraph).

Limits on New Physics (EFT) and Unitarity

- **95% CL limits** set on 9 (8) dimension-8 operators without (with) consideration of tree-level unitarity violation.
- These constraints are similar to those obtained by the CMS Collaboration.



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Thank you for listening!

Any questions?